(1). The system of linear equations

$$3x + 2y = 10$$
$$6x + 4y = 15$$

has no solution.

If we choose right hand side 20 then the system of equations has infinitely many solutions.

$$3x + 2y = 10$$
$$6x + 4y = 20$$

Let $y = c \in \mathbb{R}$. Then $x = \frac{10 - 2c}{3}$

So, the solutions are given by,

$x = \frac{10-2c}{3}$, $y = c$, c be any real number.

(2) Let A be a lower ~~triangular~~ triangular matrix.

and $A = \begin{bmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ a_{21} & a_{22} & 0 & \cdots & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{n1} & a_{n2} & 0 & \cdots & a_{nn} \end{bmatrix}$

A will be invertible iff $\det(A) \neq 0$

Now, $\det(A) = a_{11} a_{22} \cdots a_{nn} = \prod_{i=1}^{n} a_{ii}$

So, $\det(A) \neq 0 \iff a_{ii} \neq 0 \ \forall i$

i.e. A is invertible iff each diagonal entry of A is non-zero.

For a upper ~~[illegible]~~ triangular matrix the proof is also similar.

Hence a triangular matrix A is invertible iff each diagonal entry of A is non-zero.

(3) Given $A_1, A_2, \cdots, A_r$ are invertible matrices.

Then $\det(A_i) \neq 0$ for $i = 1, 2, \cdots, r$.

Now, $\det(A_1 A_2 \cdots A_r)$

$= \det A_1 \cdot \det A_2 \cdot \cdots \det A_r$

$\neq 0$

Thus the matrix $A_1 A_2 \cdots A_r$ is invertible.

(4) Taking $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, we get

$$A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = B \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

~~given~~ $\Rightarrow$ first column of A = first column of B

Again, taking $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, we get,

$$A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = B \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$\Rightarrow$ second column of A = second column of B.

So, A and B are two 2×2 matrices whose columns are identical.

Therefore, $A = B$.

(5) The first matrix is a row reduced echelon matrix.

The second matrix $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is not a row reduced echelon matrix, as all the non-zero rows should be above zero rows.

The third matrix, $\begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$ is not of row reduced echelon form.

In the third row all the elements ~~in~~ above leading 1 should be zero.

The fourth matrix $\begin{bmatrix} 1 & 0 & 3 \\ 0 & 2 & 7 \\ 0 & 0 & 0 \end{bmatrix}$ is not of row reduced echelon form. In every non zero row the leading element should be 1, but in second row it is 2.

⑥

(a)

$$x+y+z+w=0$$
$$x-y+z+w=0$$
$$-x+y+3z+3w=0$$

writting in Augmented form,

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 1 & -1 & 1 & 1 & 0 \\ -1 & 1 & 3 & 3 & 0 \end{array}\right] \xrightarrow[R_3 \to R_1+R_2]{R_2 \to R_1 - R_2} \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 2 & 4 & 4 & 0 \end{array}\right]$$

$$\xrightarrow[R_3 \to R_2 - R_3]{} \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & -4 & -4 & 0 \end{array}\right]$$

Now by back substitution, we see that

$$-4(z+w)=0 \Rightarrow z=-w$$
$$2y=0 \Rightarrow y=0$$

and $x+y+z+w=0 \Rightarrow x+y+z-z=0$

$$\Rightarrow x=0$$

So the solution is, $x=0, y=0, z=-w, w=w$ $\{w \in \mathbb{R}$ be the free variable

(b) $x+2y+3z=1$, $x+3y+2z=1$

Augmented matrix,

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 1 & 3 & 2 & 1 \end{array}\right] \xrightarrow{R_2 \to R_1 - R_2} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & -1 & 1 & 0 \end{array}\right]$$

So,

$$-y+z=0 \Rightarrow y=z$$

and $x+2y+3z=1 \Rightarrow x+2z+3z=0 \Rightarrow x=-5z$

So the solution is $x=-5z$, $y=z$ $\{z$ is ~~arbitrary~~ free variable$\}$

(c) $x+y+z=3, \quad x+y-z=1, \quad x+y+7z=6$

Augmented matrix,

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 7 & 6 \end{array}\right] \xrightarrow[R_3 \to R_1 - R_3]{R_2 \to R_1 - R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & -6 & -3 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_2 + \frac{1}{3} R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

Since rank $A = 2$ and rank $(A|b) = 3$ $\Rightarrow$ No solution.

(d) Augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 4 & 6 \end{array}\right] \xrightarrow[R_3 \to R_1 - R_3]{R_2 \to R_1 - R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & -3 & -3 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_2 + \frac{1}{3} R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

So the solution is, $2z = 2 \Rightarrow z = 1$ and $x+y+z=3$

$\Rightarrow x+y=2 \Rightarrow x = 2-y$

Hence $x = 2-y, \; y = y, \; z = 1$ $\{y$ is arbitrary$\}$.

(e) Augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 4 & 6 \\ 1 & 1 & -4 & -1 \end{array}\right] \xrightarrow[R_4 \to R_1 - R_4]{\substack{R_2 \to R_1 - R_2 \\ R_3 \to R_1 - R_3}} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & -3 & -3 \\ 0 & 0 & 5 & 4 \end{array}\right]$$

$$\xrightarrow{R_4 \to R_2 - \frac{2}{5} R_4} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & -3 & -3 \\ 0 & 0 & 0 & 2/5 \end{array}\right] \Rightarrow \text{rank}(A) \neq \text{rank}(A|b)$$

$\Rightarrow$ No solution.

(7)

$$A = \begin{bmatrix} 4 & 0 & 2 & 0 & -3 \\ 4 & 1 & 0 & -1 & 0 \\ 0 & 3 & 1 & 0 & -2 \\ 8 & 2 & 0 & -2 & 0 \\ 0 & 0 & 1 & 0 & -3 \end{bmatrix} \xrightarrow[R_4 \to R_4 - 2R_1]{R_2 \to R_2 - R_1} \begin{bmatrix} 4 & 0 & 2 & 0 & -3 \\ 0 & 1 & -2 & -1 & 3 \\ 0 & 3 & 1 & 0 & -2 \\ 0 & 2 & -4 & -2 & 6 \\ 0 & 0 & 1 & 0 & -3 \end{bmatrix}$$

$$\xrightarrow[R_4 \to R_4 - 2R_2]{R_3 \to R_3 - 3R_2} \begin{bmatrix} 4 & 0 & 2 & 0 & -3 \\ 0 & 1 & -2 & -1 & 3 \\ 0 & 0 & 7 & 3 & -11 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -3 \end{bmatrix}$$

$$\xrightarrow{R_3 \leftrightarrow R_4} \begin{bmatrix} 4 & 0 & 2 & 0 & -3 \\ 0 & 1 & -2 & -1 & 3 \\ 0 & 0 & 7 & 3 & -11 \\ 0 & 0 & 1 & 0 & -3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_4 \to R_3 - 7R_4} \begin{bmatrix} 4 & 0 & 2 & 0 & -3 \\ 0 & 1 & -2 & -1 & 3 \\ 0 & 0 & 7 & 3 & -11 \\ 0 & 0 & 0 & 3 & 10 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{R_3 \to R_3 - R_4} \begin{bmatrix} 4 & 0 & 2 & 0 & -3 \\ 0 & 1 & -2 & -1 & 3 \\ 0 & 0 & 7 & 0 & -21 \\ 0 & 0 & 0 & 3 & 10 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[R_3 \to \frac{1}{7} R_3,\ R_4 \to \frac{1}{3} R_4]{R_1 \to \frac{1}{4} R_1} \begin{bmatrix} 1 & 0 & 1/2 & 0 & -3/2 \\ 0 & 1 & -2 & -1 & 3 \\ 0 & 0 & 1 & 0 & -3 \\ 0 & 0 & 0 & 1 & \frac{10}{3} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{R_2 \to R_2 + 2R_3 + R_4} \begin{bmatrix} 1 & 0 & 1/2 & 0 & -3/2 \\ 0 & 1 & 0 & 0 & 1/3 \\ 0 & 0 & 1 & 0 & -3 \\ 0 & 0 & 0 & 1 & 10/3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{R_1 \to R_1 - \frac{1}{2} R_3} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1/3 \\ 0 & 0 & 1 & 0 & -3 \\ 0 & 0 & 0 & 1 & 10/3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Answer

⑧ Finding inverse using Gauss - Jordan Method

ⓐ
$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

Adjoining the identity matrix to the right side of A.

$$\left[\begin{array}{ccc|ccc} 2 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow[R_3 \to R_1 - 2R_3]{R_2 \to R_1 - 2R_2} \left[\begin{array}{ccc|ccc} 2 & 1 & 1 & 1 & 0 & 0 \\ 0 & -3 & -1 & 1 & -2 & 0 \\ 0 & -1 & -3 & 1 & 0 & -2 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_2 - 3R_3} \left[\begin{array}{ccc|ccc} 2 & 1 & 1 & 1 & 0 & 0 \\ 0 & -3 & -1 & 1 & -2 & 0 \\ 0 & 0 & 8 & -2 & -2 & 6 \end{array}\right]$$

$$\xrightarrow{R_2 \to R_2 + \frac{1}{8}R_3} \left[\begin{array}{ccc|ccc} 2 & 1 & 1 & 1 & 0 & 0 \\ 0 & -3 & 0 & \frac{3}{4} & -\frac{9}{4} & -\frac{3}{4} \\ 0 & 0 & 8 & -2 & -2 & 6 \end{array}\right]$$

$$\xrightarrow[\substack{R_2 \to -\frac{1}{3}R_2 \\ R_3 \to \frac{1}{8}R_3}]{R_1 \to \frac{1}{2}R_1} \left[\begin{array}{ccc|ccc} 1 & \frac{1}{2} & \frac{1}{2} & 1/2 & 0 & 0 \\ 0 & 1 & 0 & -\frac{1}{4} & \frac{3}{4} & \frac{1}{4} \\ 0 & 0 & 1 & -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - \frac{1}{2}R_2 - \frac{1}{2}R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{12}{16} & -\frac{12}{48} & -\frac{12}{48} \\ 0 & 1 & 0 & -\frac{12}{48} & \frac{36}{48} & -\frac{12}{48} \\ 0 & 0 & 1 & -\frac{12}{48} & -\frac{12}{48} & \frac{36}{48} \end{array}\right]$$

So,

$$A^{-1} = \begin{bmatrix} 0.75 & -0.25 & -0.25 \\ -0.25 & 0.75 & -0.25 \\ -0.25 & -0.25 & 0.75 \end{bmatrix}$$

Answer

(b)

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \\ 2 & 4 & 7 \end{bmatrix}$$

Then

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 1 & 3 & 2 & 0 & 1 & 0 \\ 2 & 4 & 7 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow[R_3 \to R_3 - 2R_1]{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & 1 & -2 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - 2R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 5 & 3 & -2 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & 1 & -2 & 0 & 1 \end{array}\right]$$

$$\xrightarrow[R_2 \to R_2 + R_3]{R_1 \to R_1 - 5R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 13 & -2 & -5 \\ 0 & 1 & 0 & -3 & 1 & 1 \\ 0 & 0 & 1 & -2 & 0 & 1 \end{array}\right]$$

So

$$B^{-1} = \begin{bmatrix} 13 & -2 & -5 \\ -3 & 1 & 1 \\ -2 & 0 & 1 \end{bmatrix}$$

(c)

$$C = \begin{bmatrix} 2 & -1 & 3 \\ -1 & 3 & -2 \\ 2 & 4 & 1 \end{bmatrix}$$

then

$$\left[\begin{array}{ccc|ccc} 2 & -1 & 3 & 1 & 0 & 0 \\ -1 & 3 & -2 & 0 & 1 & 0 \\ 2 & 4 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_1 \to \frac{R_1}{2}} \left[\begin{array}{ccc|ccc} 1 & -0.5 & 1.5 & 0.5 & 0 & 0 \\ -1 & 3 & -2 & 0 & 1 & 0 \\ 2 & 4 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow[R_3 \to R_3 - 2R_1]{R_2 \to R_2 + R_1} \left[\begin{array}{ccc|ccc} 1 & -0.5 & 1.5 & 0.5 & 0 & 0 \\ 0 & 2.5 & -0.5 & 0.5 & 1 & 0 \\ 0 & 5 & -2 & -1 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \to \frac{R_2}{2.5}} \left[\begin{array}{ccc|ccc} 1 & -0.5 & 1.5 & 0.5 & 0 & 0 \\ 0 & 1 & -0.2 & 0.2 & 0.4 & 0 \\ 0 & 5 & -2 & -1 & 0 & 1 \end{array}\right]$$

$$\xrightarrow[R_3 \to R_3 - 5R_2]{R_1 \to R_1 + \frac{1}{2}R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 1.4 & 0.6 & 0.2 & 0 \\ 0 & 1 & -0.2 & 0.2 & 0.4 & 0 \\ 0 & 0 & -1 & -2 & -2 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \to -R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 1.4 & 0.6 & 0.2 & 0 \\ 0 & 1 & -0.2 & 0.2 & 0.4 & 0 \\ 0 & 0 & 1 & 2 & 2 & -1 \end{array}\right]$$

$$\xrightarrow[R_2 \to R_2 + 0.2R_3]{R_1 \to R_1 - 1.4R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -2.2 & -2.6 & 1.4 \\ 0 & 1 & 0 & 0.6 & 0.8 & -0.2 \\ 0 & 0 & 1 & 2 & 2 & -1 \end{array}\right]$$

So $$C^{-1} = \begin{bmatrix} -2.2 & -2.6 & 1.4 \\ 0.6 & 0.8 & -0.2 \\ 2 & 2 & -1 \end{bmatrix}$$

(9) $a, b \in \mathbb{R}$ Given equations,

$$\begin{aligned} 2x + y + az &= 1 \\ x + z &= 2 \\ x + y - z &= b \end{aligned}$$

(a) Augmented matrix,

$$\left[\begin{array}{ccc|c} 2 & 1 & a & 1 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -1 & b \end{array}\right]$$

$$\xrightarrow{R_2 \leftrightarrow R_1} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 2 & 1 & a & 1 \\ 1 & 1 & -1 & b \end{array}\right]$$

$$\xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & a-1 & -3 \\ 0 & 1 & -2 & b-2 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & a-1 & -3 \\ 0 & 0 & -(a+1) & b+1 \end{array}\right] \xrightarrow{R_3 \to \frac{-1}{a+1} R_3} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & a-1 & -3 \\ 0 & 0 & 1 & -\frac{b+1}{a+1} \end{array}\right]$$

if $a \neq -1$

If $a = -1$, then this is the RREF form.

$$\xrightarrow[R_2 \to R_2 - (a-1) R_3]{R_1 \to R_1 - R_3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 + \frac{b+1}{a+1} \\ 0 & 1 & 0 & -3 + \frac{(b+1)(a-1)}{a+1} \\ 0 & 0 & 1 & -\left(\frac{b+1}{a+1}\right) \end{array}\right]$$

Answer

(b) (1) No solution, if $a = -1$ and $b \neq -1$

(2) A unique solution if $a \neq -1$

(3) Infinitely many solution if $a = -1$ and $b = -1$.

(c) If $a \neq -1$, $x = 2 + \frac{b+1}{a+1}$, $y = -3 + \frac{(b+1)(a-1)}{a+1}$, $z = -\frac{b-1}{a+1}$

If $a = -1$ and $b = -1$, $x = 2$, $y = -3$ and z is arbitrary.

(10) By definition of elementary matrix, we know that.
Elementary matrix is a matrix which differs from the identity matrix by one single elementary row operation. So each elementary matrix is row reduced equivalance to identity matrix. Hence, each elementary matrix is invertible.
Yes, inverse of an elementary matrix is also an elementary matrix.

(11) Let $A \in M_{m \times n}(\mathbb{R})$ and $b \in \mathbb{R}^m$

Let $S \subseteq \mathbb{R}^n$ is the solution's of $AX = 0$.

and X_0 be the solution of $AX = b$.

$$X_0 + S = \{ X_0 + V : V \in S \}$$

Let $Y \in X_0 + S$

i.e. $Y = X_0 + V$ for some $V \in S$.

$$AY = A(X_0 + V) = AX_0 + AV$$
$$= b + 0 = b$$

$\therefore$ Y is a solution of $AX = b$.

i.e. every vector in $\underline{X_0 + S}$ is a solution of $AX = b$

If Y is a solution of $AX = b$

i.e. $AY = b$

Let $Y = Y + (X_0 - X_0) = X_0 + (Y - X_0)$

$$A(Y - X_0) = AY - AX_0 = b - b = 0$$

$$\Rightarrow Y - X_0 \in S$$

$$\therefore Y = X_0 + (Y - X_0) \in \underline{X_0 + S}$$

$\therefore$ i.e. If Y is a solution of $AX = b$

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then $Y \in X_0 + S$.

$\therefore$ Every solution of $AX = b$ is given by

$$X_0 + S$$

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(12). Let A is an $n \times n$ matrix

$\underline{\underline{(a) \Rightarrow (b)}}$

Let A is invertible $\Rightarrow A^{-1}$ exists

for any $b \in \mathbb{R}^n$,

Consider the system of equation $Ax = b$

$$A^{-1}(Ax) = A^{-1} b$$

$$\Rightarrow \quad \underline{\underline{x = A^{-1} b}} \in \mathbb{R}^n$$

$\therefore$ $x_0 = A^{-1} b$ is a solution of $Ax = b$.

If Y is any solution of $Ax = b$

$$\Rightarrow \quad AY = b$$

$$\Rightarrow \quad Y = A^{-1} b = x_0 \quad \text{ie} \quad Y = x_0$$

$\therefore$ Solution is $\underline{\underline{\text{unique}}}$

ie $x_0 = A^{-1} b$ is $\underline{\underline{\text{the unique}}}$ solution of $Ax = b$

$\underline{\underline{(b) \Rightarrow (a)}}$

Given that for any $b \in \mathbb{R}^n$, $Ax = b$ has a unique solution.

$\Rightarrow$ every $b \in \mathbb{R}^n$ is $C(A)$

$$\Rightarrow \quad C(A) = \mathbb{R}^n$$

$$\Rightarrow \quad \text{Rank}(A) = n \Rightarrow A \text{ is invertible}$$

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(13)

(a) we know that $\text{Rank}(A) = \dim(C(A))$

where $C(A)$ - columnspace of A.

we have $\dim(U+W) = \dim(U) + \dim(W) - \dim(U \cap W)$.

let $U = C(A)$ and $W = C(B)$.

$$\dim(C(A) + C(B)) = \dim(C(A)) + \dim(C(B)) - \dim(C(A) \cap C(B))$$

$$\Rightarrow \dim(C(A) + C(B)) \leq \dim(C(A)) + \dim(C(B)) \quad \text{—} (1)$$

$$\left[\because \dim(C(A) \cap C(B)) \geq 0.\right]$$

we know that, If A is $m \times n$ matrix, then

$$C(A) = \{AX : X \in \mathbb{R}^{n \times 1}\}$$

If $Y \in C(A+B) \Rightarrow Y = (A+B)X$

$$= AX + BX$$

$$\Rightarrow Y \in C(A) + C(B)$$

$$\left(\because AX \in C(A) \text{ \& } BX \in C(B)\right)$$

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$\therefore \quad C(A+B) \subseteq C(A) + C(B)$

$\Rightarrow \quad \dim(C(A+B)) \leq \dim((C(A)) + (C(B))) \quad \text{—} (2)$

(1) & (2)

$\Rightarrow \quad \dim(C(A+B)) \leq \dim(C(A)) + \dim(C(B))$

$\Rightarrow \quad \text{Rank}(A+B) \leq \text{Rank}(A) + \text{Rank}(B)$

(b). Let AB is defined.

Let $Y \in C(AB)$

$\Rightarrow Y = (AB)X$

$\Rightarrow Y = A(BX)$

$\Rightarrow Y \in C(A)$

$\therefore \quad C(AB) \subseteq C(A)$

$\Rightarrow \quad \text{Rank}(AB) \leq \text{Rank}(A)$

Since each Row of (AB) is a linear combination of row's of B

$\Rightarrow \quad \text{Row space}(AB) \subseteq \text{Row space}(B)$

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$$\Rightarrow \dim(\text{Row space}(AB)) \leq \dim(\text{rowspace}(B))$$

$$\Rightarrow \quad \text{Rank}(AB) \leq \text{Rank}(B)$$

(14). Let $A, B, C \in M_{m\times n}(\mathbb{R})$.

(i) <u>Closure property.</u>

we know that sum of two $m\times n$ matrix is again an $m\times n$ matrix.

$$\therefore \quad A+B \in M_{m\times n}(\mathbb{R}).$$

(ii) <u>Associative property</u>

Since Matrix addition is associative,

$$(A+B)+C = A+(B+C).$$

(iii) <u>Commutative property.</u>

Since matrix addition is commutative

$$A+B = B+A.$$

(iv) let $O = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}_{m\times n} \in M_{m\times n}(\mathbb{R})$

and $A+O = O+A = A.$

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$\therefore$ zero vector in $M_{m\times n}(\mathbb{R})$ is $O = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & & & \\ 0 & 0 & \cdots & 0 \end{bmatrix}_{m\times n}$

(v) <u>Additive inverse</u>

If $A \in M_{m\times n}(\mathbb{R})$, then $-A \in M_{m\times n}(\mathbb{R})$

and $A + (-A) = O$

Let c be a scalar.

(vi) for any $A \in M_{m\times n}(\mathbb{R})$, $cA \in M_{m\times n}(\mathbb{R})$

(vii) $c(A+B) = c\cdot A + cB$

(viii) $(c_1 + c_2)A = c_1 A + c_2 A$ where c_1 & c_2 are scalars

(ix) $c_1(c_2 A) = (c_1 c_2) A$

(x) $1\cdot A = A$ for any $A \in M_{m\times n}(\mathbb{R})$

$\therefore$ $M_{m\times n}(\mathbb{R})$ is a vector space over matrix addition and scalar multiplication.

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(15) In $\mathbb{R}$, $x \oplus y = x+y-1$

$$a \odot x = a(x-1)+1.$$

Let $x, y, z \in \mathbb{R}$ and λ, μ be scalars

(i) Closure property

$$x \oplus y = x+y-1 \in \mathbb{R}. \quad \because x, y \in \mathbb{R}.$$

(ii) Associative property

$$(x \oplus y) \oplus z = (x+y-1) \oplus z$$
$$= x+y-1+z-1$$
$$= \underline{\underline{x+y+z-2}}$$

$$x \oplus (y \oplus z) = x \oplus (y+z-1)$$
$$= x+y+z-1-1$$
$$= \underline{\underline{x+y+z-2}}$$

$$\therefore \underline{\underline{x \oplus (y \oplus z) = (x \oplus y) \oplus z}}$$

(iii) ~~As~~ Commutative property

$$x \oplus y = x+y-1 = y \oplus x.$$

(iv) Existence of additive identity.

Let 'e' is the additive identity,

i.e. $x \oplus e = x$

$$\Rightarrow x+e-1 = x \Rightarrow \underline{\underline{e=1}}.$$

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$\therefore$ $e = 1$ is the additive identity.

for any $x \in \mathbb{R}$,

$$x \oplus 1 = x + 1 - 1 = \underline{\underline{x}}$$

(v) <u>Additive inverse</u>

Let x^{-1} is the additive inverse of x

ie $x \oplus x^{-1} = 1$

$$x + x^{-1} - 1 = 1$$

$$\Rightarrow \quad x^{-1} = 2 - x \in \mathbb{R} \qquad (\because x \in \mathbb{R})$$

so for any $x \in \mathbb{R}$, $\underline{\underline{2-x}}$ is the additive inverse

$$x \oplus 2 - x = x + 2 - x - 1 = 1 \quad \forall x \in \mathbb{R}$$

(vi) $1 \odot x = 1(x-1) + 1$

$$= x - 1 + 1 = x$$

ie $1 \odot x = x \quad \forall x \in \mathbb{R}$

(vii) $\lambda \odot (x \oplus y) = \lambda \odot (x + y - 1)$

$$= \lambda(x + y - 2) + 1$$

$$(\lambda \odot x) \oplus (\lambda \odot y) = [\lambda(x-1) + 1] \oplus [\lambda(y-1) + 1]$$

$$= \lambda(x-1) + 1 + \lambda(y-1) + 1 - 1$$

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$$= \lambda x - \lambda + 1 + \lambda y - \lambda + 1 - 1$$

$$= \lambda x - \lambda + \lambda y - \lambda + 1$$

$$= \lambda (x + y - 2) + 1$$

$$\therefore \quad \lambda \odot (x \oplus y) = (\lambda \odot x) \oplus (\lambda \odot y)$$

(viii) $(\lambda + \mu) \odot x = (\lambda + \mu)(x - 1) + 1$

$$= \lambda x + \mu x - \lambda - \mu + 1$$

$$(\lambda \odot x) \oplus (\mu \odot x) = (\lambda (x-1) + 1) \oplus (\mu (x-1) + 1)$$

$$= \lambda (x-1) + 1 + \mu (x-1) + 1 - 1$$

$$= \lambda x - \lambda + 1 + \mu x - \mu$$

$$= \lambda x + \mu x - \lambda - \mu - 1$$

$$\therefore \quad (\lambda + \mu) \odot x = (\lambda \odot x) \oplus (\mu \odot x)$$

(ix) $(\lambda \mu) \odot x = \lambda \mu (x - 1) + 1$

$$\lambda \odot (\mu \odot x) = \lambda \odot (\mu (x-1) + 1)$$

$$= \lambda (\mu (x-1) + 1 - 1) + 1 = \underline{\lambda \mu (x-1) + 1}$$

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$\therefore (\lambda\mu) \odot x = \lambda \odot (\mu \odot x)$

$\therefore$ $\mathbb{R}$ is a vector space with the operations

$$x \oplus y = x + y - 1$$

$$a \odot x = a(x-1) + 1$$

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